Naltrexone: Treatments for Alcohol Dependence

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Problem Definition & Motivation

Currently, ~6% of American adults suffer from alcohol use disorder (AUD). ***Naltrexone*** is a FDA approved drug to combat AUD with two administration types:

- **Oral** (50 mg / day)
- **Injection** (380 / 4 weeks)

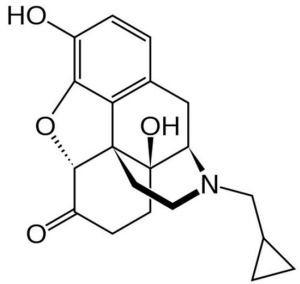
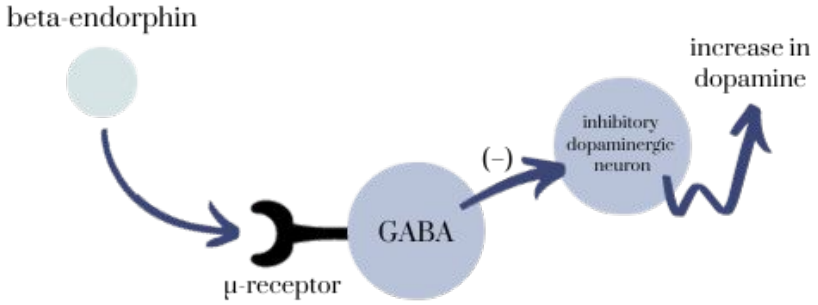


Figure 1:
Structure of Naltrexone

Using **pharmacokinetic modeling**, we can evaluate:

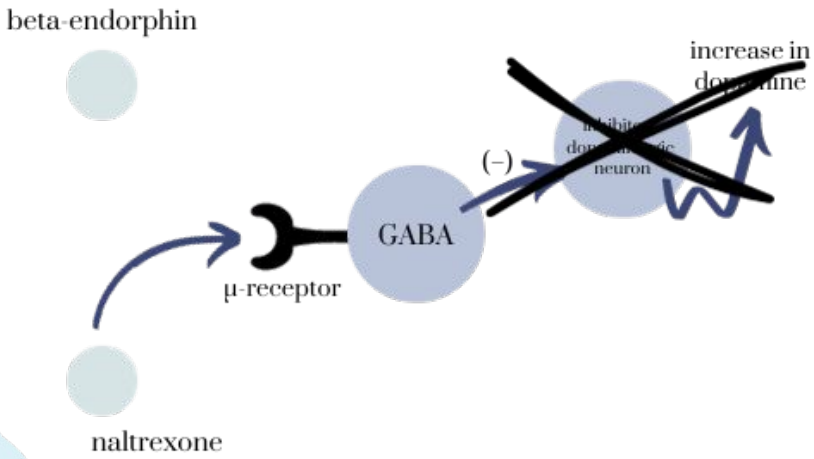
- Drug concentrations and rates between different administration routes
- Naltrexone's physiological impact on the disease state
- Effects of missing a dosage

Mechanisms: Alcohol Dependency & Naltrexone



- Ethanol consumption releases beta endorphins (along with other endogenous opioids)
- Beta endorphins binding to μ opiate receptors removes inhibition

Mechanisms: Alcohol Dependency & Naltrexone



- Naltrexone binds to and blocks μ-receptors, competing with beta endorphins
- GABA inhibitory function retained, dopamine release limited

Overall Assumptions

1) The patient taking the dose has no other unrelated **health issues,** nor are they taking **other drugs** simultaneously
2) The patient is **not consuming alcohol** during the treatment.
3) Naltrexone **targets only μ-receptors in the brain.**
4) **μ-receptors in the brain** only bind to naltrexone/its metabolite.

Model Development - Oral

Developed ***two*** 3-compartment models:

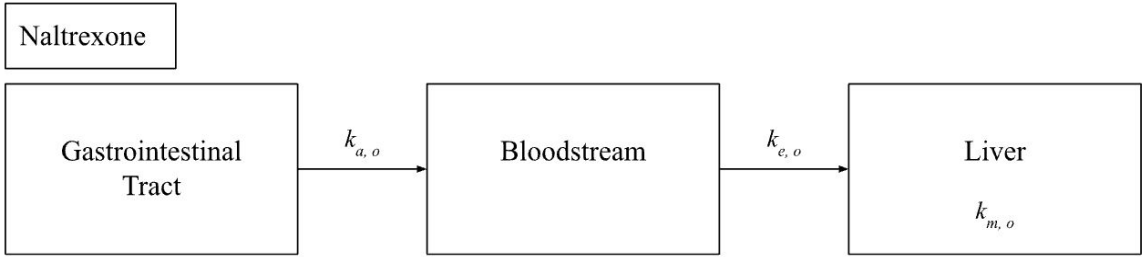
Naltrexone, drug

- Undergoes **first-pass metabolism** in liver — assume negligible physiological effect
- Assumptions:
  - Naltrexone is fully metabolized to 6B-Naltrexol

6B-Naltrexol, active metabolite

- Predominant **driver of physiological effects**
- Assumptions:
  - 6B-Naltrexol only enters the brain when binding to μ-receptor
  - Unbound 6B-Naltrexol remains in bloodstream

Conceptual Model - Oral

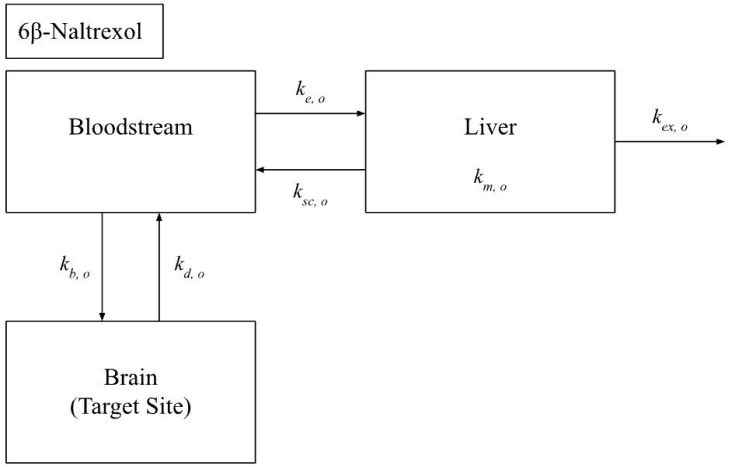


$$\frac{dD_{GI}}{dt} = -k_{a,o}D_{GI}$$

$$\frac{dD_{BS,o}}{dt} = k_{a,o}D_{GI} - k_{e,o}D_{BS,o}$$

$$\frac{dD_{L,o}}{dt} = k_{e,o}D_{BS,o} - k_{sc,o}D_{L,o} - k_{m,o}D_{L,o}$$

Conceptual Model - Oral



$$\frac{dD_{M,BS}}{dt} = -k_{e,o}D_{M,BS} - k_{sc,o}D_{M,L} - k_{b,o}D_{M,BS}D_{\mu,o} - k_{d,o}D_{M,B}$$

$$\frac{dD_{M,B}}{dt} = k_{b,o}D_{M,BS}D_{\mu,o} - k_{d,o}D_{M,B} - k_{deg,o}$$

$$\frac{dD_{M,L}}{dt} = k_{m,o}D_{L,o} - k_{e,o}D_{M,BS} - k_{sc,o}D_{M,L} - k_{ex,o}D_{M,L}$$

$$\frac{dD_{\mu,o}}{dt} = k_{d,o}D_{B,o} - k_{b,o}D_{BS,o}D_{\mu,o} + k_{deg,o}$$

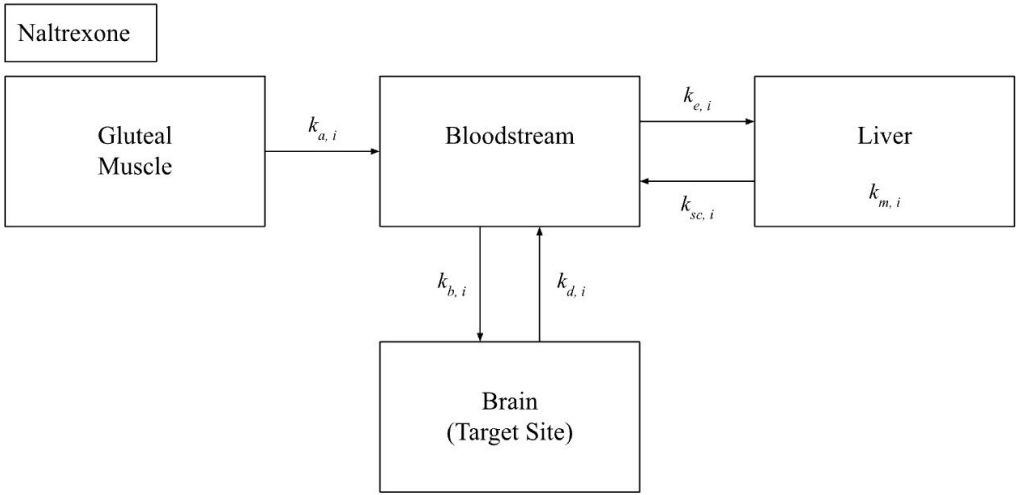
Model Development - Injection

Developed ***single*** compartmental model: **Naltrexone**

Naltrexone Component

- Direct administration into bloodstream avoids first-pass metabolism
- Assumptions
  - **Naltrexone**, not 6B-Naltrexol, is responsible for producing physiological effects in the brain
  - 6B-Naltrexol is **excreted** from liver through kidneys

Conceptual Model - Injection



$$\frac{dD_{GM}}{dt} = -k_{a,i}D_{GM}$$

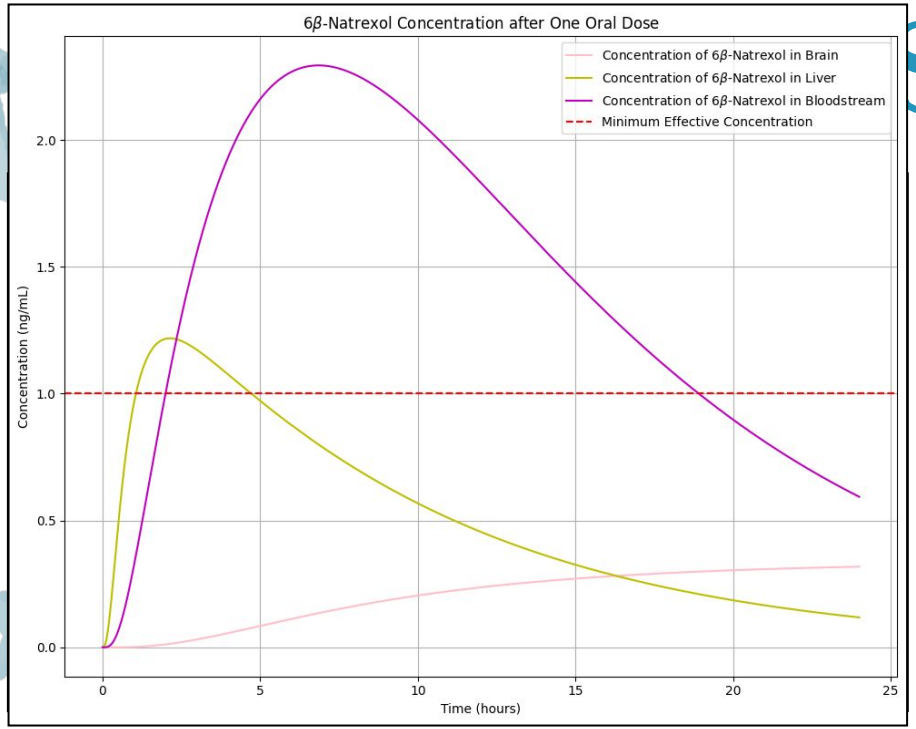
$$\frac{dD_{BS,i}}{dt} = k_{a,i}D_{GM} + k_{d,i}D_{B,i} - k_{e,i}D_{BS,i} - k_{b,i}D_{BS,i}D_{\mu,i}$$

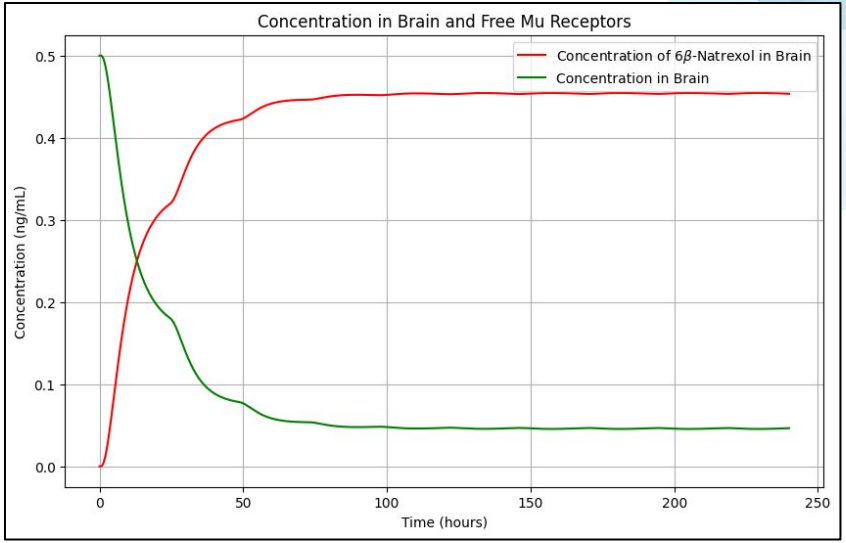
$$\frac{dD_{B,i}}{dt} = k_{b,i}D_{BS,i}D_{\mu,i} - k_{d,i}D_{B,i} - k_{deg}$$

$$\frac{dD_{\mu,i}}{dt} = k_{d,i}D_{B,i} - k_{b,i}D_{BS,i}D_{\mu,i} + k_{deg}$$

$$\frac{dD_{L,i}}{dt} = k_{e,i}D_{BS,i} - k_{sc,i}D_{L,i} - k_{m,i}D_{L,i}$$

Strategy - Oral

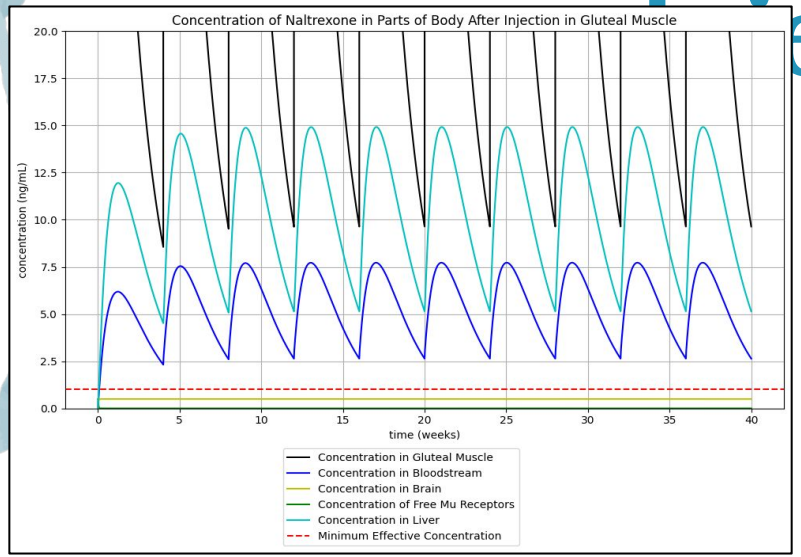


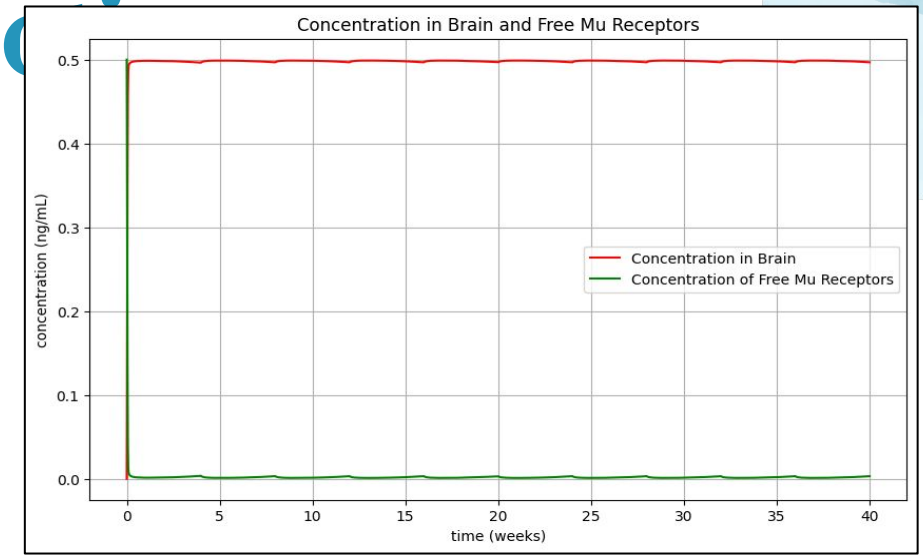


50 mg daily doses, tracking **6B-Naltrexol**

- Lower binding affinity than Naltrexone
- Steady state after 75 hours, 95% of μ-receptors blocked
- Meets minimum effective concentration in bloodstream for ~15h (pink)

Results: Dosing Strategy - Injection





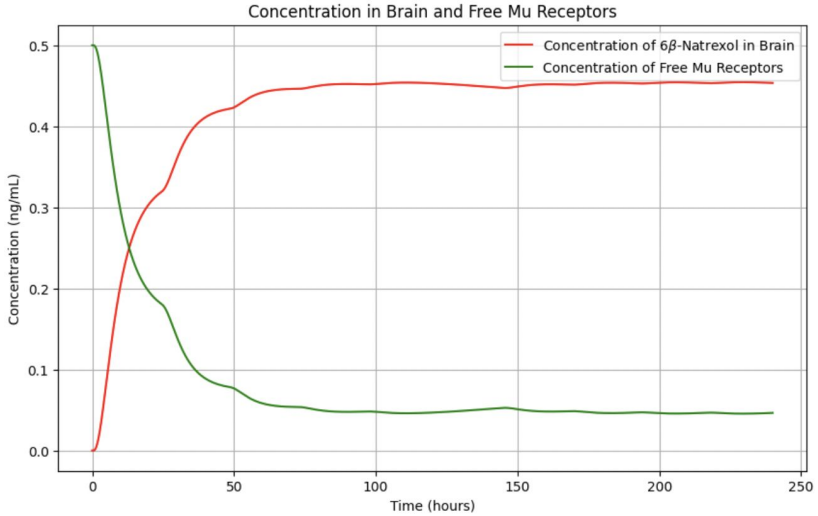
380 mg doses per 4 weeks, tracking **Naltrexone**

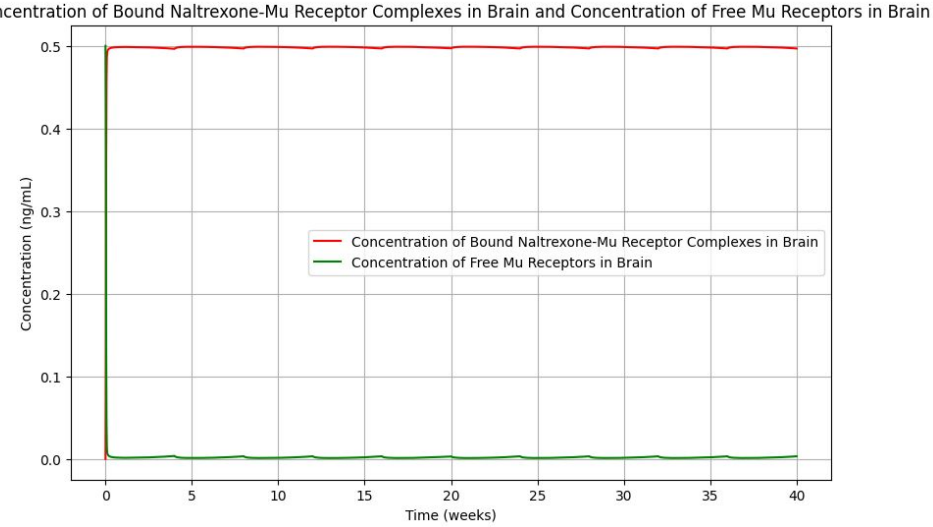
- Steady state rapidly achieved, almost 100% of μ-receptors blocked
- Bloodstream concentration (blue) above minimum effective concentration throughout 40 week period

Results: Physiological Modulation in Free μ-Receptors

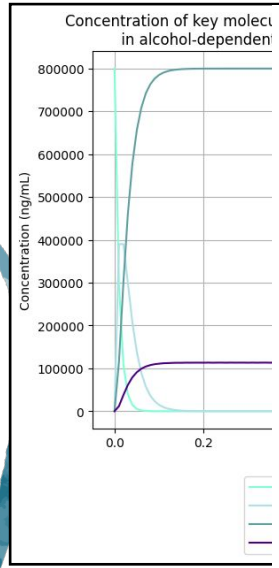
Elevated μ-receptor concentrations correlate to the phenomenon of craving alcohol in dependent patients

- **oral:** 6B-naltrexol, fluctuates around 95% blockade
- **injection:** naltrexone, nearly 100% blockade, sustained





Results: Physiological Modulation in Dopamine

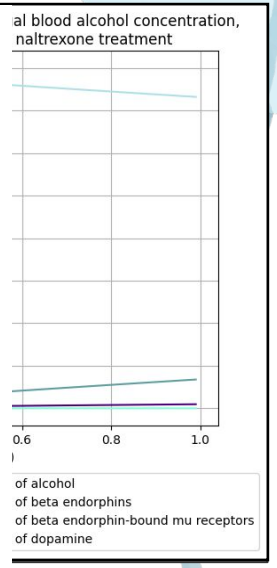


$$\frac{dD_{alc}}{dt} = -k_1 D_{alc}$$

$$\frac{dD_{beta}}{dt} = k_1 D_{alc} - k_2 D_{beta} D_{\mu}$$

$$\frac{dD_{beta-\mu\,complex}}{dt} = k_2 D_{beta} D_{\mu}$$

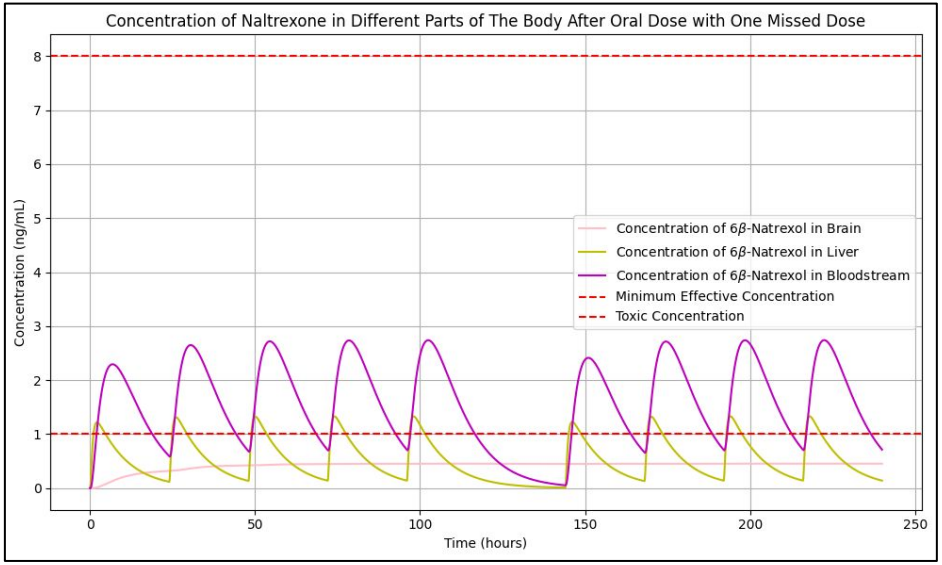
$$\frac{dD_{dopamine}}{dt} = k_3 D_{beta-\mu\,complex}$$

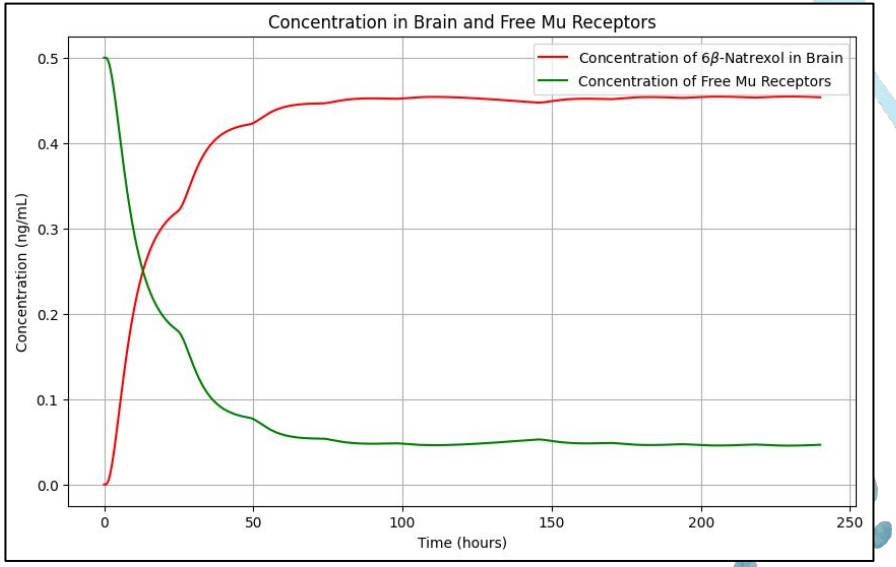


Dopami ; alcohol

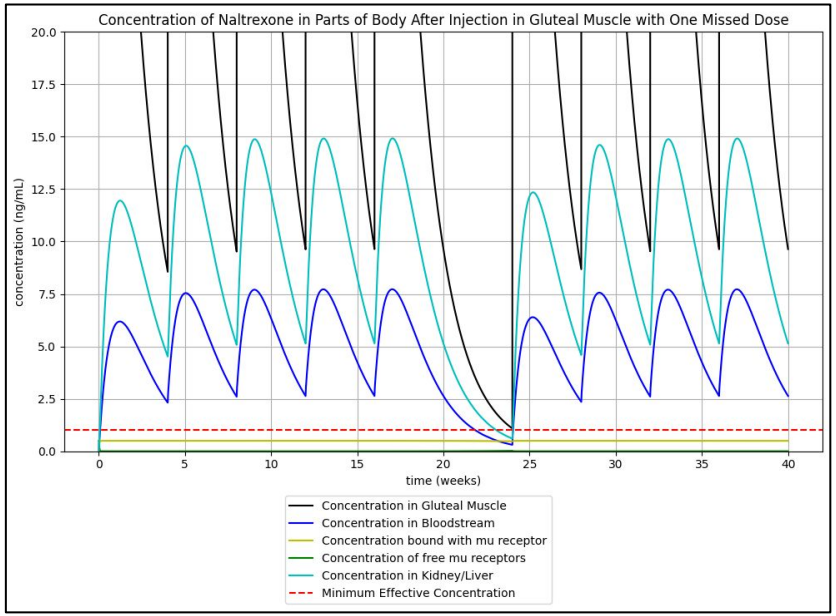
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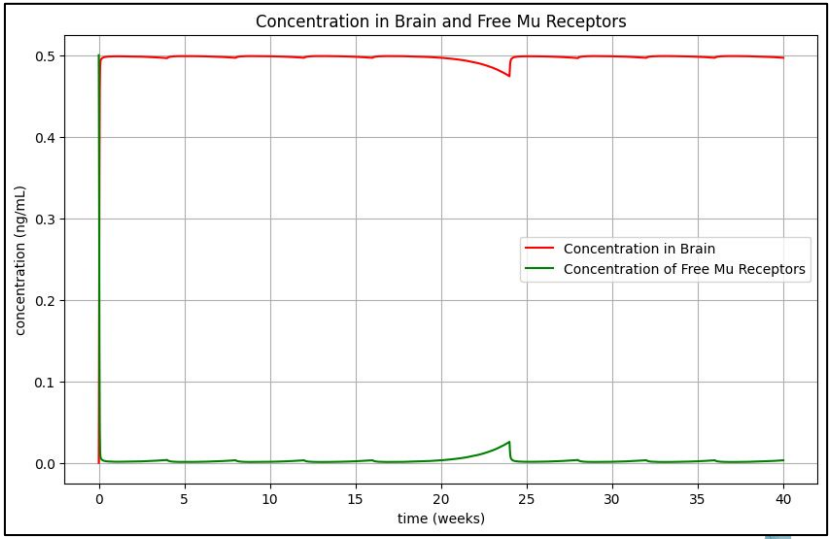
Results: Missing Oral Dose





Results: Missing Injection Dose





Cost & Ethics

REVIA (Oral)

- Cost*:
  - ~$110 per month
  - ~$1,400 per year
- More Accessible
- More Freedom/Less Invasive

VIVITROL (Injection)

- Cost*:
  - ~$1,000-1,500 per month
  - ~$13,000-20,000 per year
- Less Risk of Missing Dosage
- Better at Preventing Relapse

**Assuming no insurance*

Model Evaluation

Oral

- **Not consistently above minimum effective concentration** in bloodstream
- **Patient noncompliance:** side effects from missing a dose
- Slower binding to μ-receptors

Injection

- **Consistently above minimum effective concentration** in bloodstream
- Sufficient drug concentration present in the body even after a missed injection
- Less likely to reach toxic dose
- Faster binding to μ-receptors

Future Steps

- Faults/Limitations
  - **Modelling**: combining **brain and spinal cord**, **kidney and liver**, focusing purely on **μ-receptors**
  - Cannot examine **physiological treatment in isolation**: AUD requires holistic counselling and psychological support
- For the Future:
  - Refine models to incorporate **more variables and factors**
  - Novel **implant technology** may have increased extended release profile
  - Analyze qualitative and quantitative **behavioral** outcomes and interventions

thank you!

Thank you to Dahlia and Dr. Kim for all your help, feedback, and support on PBL 1.

Q&A